QUETZALC■■TL 1

USER MANUAL

A polyphonic cybernetic synthesizer

Sound design • Step sequencing • Preset performance • MIDI

DESKTOP EDITION / WINDOWS

This guide covers the current Quetzalcoatl 1 interface and the Windows desktop packaging. Some controls and MIDI behaviors can vary by browser, operating system, audio device, or build. Treat this as a practical guide, and use the troubleshooting section when a control does not behave as expected.

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Design note: the desktop edition uses a Soundwave-inspired palette of deep navy, steel blue, cyan, amber, and phosphor mint. The interface takes inspiration from classic hardware instrument panels without being a TB-303 clone.

1. Introduction and system overview

Quetzalcoatl 1 is a browser-based synthesizer packaged as a Windows desktop application using Electron. Its sound engine runs with Web Audio technology. The desktop package gives the instrument its own application window; the core synthesizer interface remains the same family of web technologies.

Key capabilities

- Polyphonic playing with a dual-oscillator architecture and optional double mode.
- Oscillator waveforms, octave and interval choices, detuning, noise, and sub-oscillator blend.
- Resonant voltage-controlled-filter-style tone shaping, envelope controls, and LFO modulation.
- A 16-step pattern sequencer with note steps, accent, slide, tempo, swing, gate, and pattern controls.
- Built-in chorus, delay / tape-style echo, reverb, and drive controls.
- Starter presets, user patch and performance saving, JSON import/export, and visual scopes.
- MIDI features are present in the project; actual input/output availability depends on the build and the system's MIDI devices.

Important

The first desktop build is a packaged application, not a digitally signed commercial installer. Windows may show a SmartScreen or unknown-publisher warning. Only install software you downloaded from the project’s official repository and workflow artifacts.

2. Install and launch on Windows

Download the installer

- Open the project's GitHub Actions build page: https://github.com/MENTLONE/QUETZALC-TL-1/actions/runs/39817310208
- Find the Artifacts section and download Quetzalcoatl-1-Windows-Installer. GitHub downloads the artifact as a ZIP archive.
- Extract the ZIP to a folder you can find again.
- Run the .exe installer inside the extracted folder and follow the prompts. The installer lets you choose an installation directory.
- Launch Quetzalcoatl 1 from the Start menu or desktop shortcut.

First launch checklist

- Connect speakers or headphones and confirm Windows is using the desired output device.
- Click or tap the interface once if the app needs a user gesture before audio starts.
- Play a few notes using the computer keyboard or an available MIDI controller.
- Load a starter preset and check the sequencer and effects.
- If you use MIDI hardware, connect and power it before launching the app, then verify device access.

The installer artifact is a build output. Always check the current Actions run for its status and availability; artifacts can expire according to the repository workflow's retention settings.

Project links

Repository: https://github.com/MENTLONE/QUETZALC-TL-1
Desktop branch: https://github.com/MENTLONE/QUETZALC-TL-1/tree/desktop-soundwave
Build artifact page: https://github.com/MENTLONE/QUETZALC-TL-1/actions/runs/39817310208

3. First sound in 60 seconds

1. Start with a preset

Open the preset / patch area and choose a starter sound. If audio is silent, click the interface and play a note again.

2. Play a note

Use the on-screen keyboard or the computer-keyboard mapping available in the interface. If you have a MIDI keyboard connected, try it after confirming the device is recognized.

3. Shape the tone

Lower the filter cutoff for a darker sound; raise it for a brighter sound. Increase resonance carefully to emphasize the cutoff region.

4. Add movement

Try the LFO rate and depth, then add a small amount of chorus or delay. Start with low effect levels and increase them gradually.

5. Build a pattern

Start the sequencer, click an empty step to add a note, and click an active step to clear it. Adjust tempo and swing to taste.

6. Save your sound

Use the patch/performance library controls to save your setup. Export a JSON backup before making large changes.

A good first experiment

Load a bass preset, reduce cutoff, add a little resonance, then slowly sweep cutoff while a sequence plays. Add a touch of tape delay or chorus. Save a copy before experimenting so you can return to the original sound.

4. Main controls and synthesis

The control surface is arranged in parameter groups. Labels may use abbreviations such as DCO (oscillator), VCF (filter), EG (envelope generator), and LFO (low-frequency oscillator). Values are generally adjusted with knobs or sliders; move controls in small increments while listening.

Oscillators and source mix

Control group	What it does	Try this
DCO mode	Selects the oscillator operating mode; double mode can make the sound thicker.	Compare single and double mode on a sustained note.
DCO1 waveform	Chooses the primary oscillator waveform.	Compare saw and square for different harmonic character.
DCO1 octave	Moves oscillator 1 by octave steps.	Try a lower octave for bass or a higher octave for bite.
DCO2 waveform / interval	Adds a second waveform and a pitch interval relative to oscillator 1.	Use a modest interval for a wider or more harmonically complex tone.
DCO2 detune	Offsets oscillator 2 slightly in pitch.	Small detuning adds thickness; too much can sound out of tune.
Noise / sub mix	Adds noise and low-frequency weight to the source sound.	Use noise sparingly for texture; raise sub for more low end.

Filter and envelopes

Control	Function / practical use
VCF cutoff	The main brightness control. Lower settings remove more high-frequency harmonics; higher settings sound brighter.
VCF resonance	Emphasizes frequencies around the cutoff. High resonance can become sharp or whistle-like.
Key tracking / key follow	Changes how the filter response follows note pitch. Useful for keeping higher notes bright.
Filter EG intensity	Controls how strongly the filter envelope moves the cutoff after a note starts.
Attack / decay / sustain / release	Attack is the time to rise; decay is the drop after the peak; sustain is the held level; release is the fade after note-off. Separate oscillator and filter envelopes affect tone and loudness differently.
VCF drive	Adds grit or saturation to the tone. Use lower values for cleaner sounds.
LFO rate / depth	Sets the speed and strength of periodic modulation. Depending on routing, this can animate pitch or filter movement.

Visual scope

The scope selector includes WAVE, SPECTRUM, and PHASE views. Waveform view is useful for seeing amplitude over time; spectrum view helps identify harmonic content; phase view provides a different view of the signal relationship. These displays are visual aids, not precision measurement instruments.

5. The 16-step sequencer

The sequencer is designed for short repeating patterns that can be changed while playing. The exact visual position of controls can vary with window size, but the core workflow is the same.

Create and edit a pattern

- Start or stop playback with the sequencer transport control.
- Click an empty step to activate it with a note derived from the current root and scale settings.
- Click an active step to clear it. Clearing a step also removes its accent and slide flags.
- Use the root, scale, and octave controls to change the note set used when creating steps.
- Adjust BPM to set the base tempo. Swing offsets alternating steps to create a shuffled feel.
- Use gate length to change how long sequencer notes are held.
- Use the evolution or related pattern controls to explore variations where available.

Accent and slide

Accent increases a step's emphasis and can make it sound stronger or more present. Slide connects a note into the following step for a smoother pitch transition where supported. In the current interface, Shift-click is used to toggle accent and right-click is used to toggle slide; on touch devices, these gestures may not be available in the same way.

Sequencer tips

- Start with four active steps, then add one at a time.
- Try a slower tempo and a small amount of swing before adding more notes.
- Use accent on only a few steps to create rhythmic contrast.
- If a step will not clear, make sure you are clicking an active step normally rather than using an accent or slide gesture.

Note: the sequencer is a performance tool, not a full multitrack DAW. Record or capture audio in a separate application if you need a rendered recording.

6. Effects and motion

Effects add space, movement, and character after the basic tone is working. The app includes a chorus section and an effects rack with delay / tape-style echo, reverb, and drive controls. Some builds expose named effect scenes for quick setup.

Effect	Character	Practical starting point
Chorus	Thickens and widens a sound by mixing modulated copies.	Use a low-to-moderate mix on pads, brass, and sustained sounds.
Delay / Space Echo	Repeats notes after the original sound.	Start with low feedback and mix; increase until repeats support the rhythm.
Tape age / wow / flutter	Adds pitch drift and softened, worn-tape character to echoes where available.	Use subtle amounts for warmth; stronger settings create obvious instability.
Reverb	Adds a sense of room or space.	Keep bass sounds relatively dry; use more for pads and atmospheric sounds.
Drive	Adds harmonics and saturation.	Raise gradually and reduce output level if the sound becomes too loud.
Effect scenes	Named combinations such as dubby, atmospheric, tape-worn, chaotic, or bypass settings may be available.	Select a scene, listen, then adjust individual controls to taste.

Avoid clipping

Effects can increase perceived loudness and peaks. If audio distorts unexpectedly, reduce the instrument output or effect mix before raising the Windows system volume. Drive is intentional saturation, but uncontrolled digital clipping is usually not.

7. Presets and performance library

Starter presets provide a quick way to explore the instrument. The library is intended to save complete sound/performance setups, including synth parameters and supported sequencer and effect settings.

Starter sounds

Preset	Suggested use / starting idea
ACID VIPER	Resonant bass and sequenced patterns.
SUB ORBIT	Deep, weighty low-frequency parts.
GLASS CONSTELLATION	Bright, crystalline textures.
ANALOG BRASS	Punchy, harmonically rich lead or chord stabs.
NOCTURNE PAD	Slower sustained atmospheric tones.
LASER NEEDLE	Focused, bright, cutting sounds.
WARM ELECTRIC	Rounded, mellow tones.
INDUSTRIAL RITUAL	Rougher, darker, percussive textures.
TROPICAL PULSE	Rhythmic, lively sequence ideas.
CHORUS CLOUD	Wide, modulated, sustained sounds.

Saving a setup

- Dial in the sound and sequence you want.
- Use the library's save action and enter a recognizable name if prompted.
- Reload the saved entry to confirm it restores the expected sound.
- Export a JSON library or patch backup periodically, especially before replacing files or clearing browser data.

The preset names above describe the starter library known for this project. If a preset or control is absent in a particular build, use the installed interface as the source of truth.

8. Keyboard, audio, and MIDI

Computer keyboard and on-screen notes

Use the on-screen keyboard when available, or the computer-keyboard note mapping provided by the app. A key may not trigger sound until the application has received a click or other user gesture. Release notes cleanly when testing held or sustained sounds.

Audio output

- In Windows Settings, select the correct output device before launching the synth.
- Check the device's volume and mute state, then test with a simple preset and no effects.
- If sound stops after changing audio devices, close and reopen the application.
- Bluetooth audio can add noticeable latency; wired headphones or an audio interface are usually better for live playing.

MIDI controller

- Connect the MIDI device before starting the app when possible.
- Confirm Windows recognizes the device and close other software that may exclusively hold it.
- Test a note with a simple preset first, then test note release, pitch/modulation controls, and sequencer synchronization if supported.
- MIDI support can depend on the browser/Electron build and Web MIDI availability. If the device is not detected, consult the current repository issue tracker and build notes rather than assuming every MIDI feature is supported in every configuration.

> **MIDI input/output is not guaranteed for every device or build. The first Windows installer has passed the packaging workflow, but hardware compatibility should be tested on the target computer.**

9. Saving, exporting, and backups

Quetzalcoatl 1 includes local persistence and JSON import/export features in the web application. Exact storage behavior can differ between the browser edition and the packaged desktop edition, so keep your own backups.

Recommended backup routine

- Export your patches or performance library to a JSON file after a productive session.
- Store a copy outside the application folder, such as Documents or a dedicated project backup folder.
- Use clear filenames, for example: Q1-library-2026-10-22.json.
- Before importing a library, export the current library first in case you need to roll back.
- Do not assume uninstalling or reinstalling the app will preserve every locally saved item. Keep exported backups.

JSON files are data backups, not executable programs. Import only files you trust and that were exported by Quetzalcoatl 1 or are compatible with its library format.

10. Troubleshooting

The app opens but there is no sound

Click inside the app, play a note, check Windows output device and volume, and try a starter preset with effects reduced. Restart the app after changing audio devices.

Windows shows a security warning

The installer is not code-signed. Confirm you downloaded it from the official project repository's Actions artifact, then follow your organization's Windows security policy. Do not bypass warnings for files from unknown sources.

The installer or ZIP cannot be found

Open the specific successful Actions run and look under Artifacts. Artifacts expire; if it has expired, a new workflow run is needed.

MIDI keyboard is not detected

Reconnect it, check Windows device recognition, launch the app after connecting, and close competing MIDI applications. MIDI support may vary by environment.

A sequencer step will not turn off

Click an active step normally. Shift-click is reserved for accent and right-click for slide in the current interface.

Sound is too harsh or distorted

Lower drive, resonance, output level, and effect mix. Start with cutoff and oscillator levels at moderate settings.

The interface looks cramped

Maximize the app window. The minimum window size is designed for desktop use; very small windows may compress controls.

A saved patch is missing

Look for a JSON export or backup. Local browser/app storage is not a substitute for a separate backup file.

An effect or control differs from this guide

The UI may have changed between commits. Check the desktop branch and latest project notes; this manual describes the feature set targeted by the initial desktop build.

Before reporting a bug

Include your Windows version, the desktop build or commit if known, the steps to reproduce the issue, what you expected, what happened instead, and whether the same behavior occurs in the browser edition. Do not include private credentials or access tokens.

11. Quick reference and project links

Goal	Start here
Get sound quickly	Load a starter preset, click the interface, and play a note.
Make a darker tone	Lower VCF cutoff; add resonance only as needed.
Make a wider tone	Try double mode, oscillator detune, or chorus in small amounts.
Make a repeating phrase	Start sequencer, activate steps, set root/scale/octave, then adjust BPM and swing.
Make notes glide	Use slide on a step where supported; check the note transition during playback.
Save your work	Save the patch/performance in the library and export a JSON backup.
Troubleshoot sound	Check user gesture, Windows output device, volume, preset, and effect levels.

Project resources

GitHub repository
Desktop branch: desktop-soundwave
GitHub Actions builds

Quetzalcoatl 1 — make a sound, bend it, sequence it, and make it your own.

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